

Lab 4 - Wazuh SIEM / Log Analysis

Windows Authentication Event Collection and Investigation

Lab Type	SOC / SIEM Investigation
SIEM Platform	Wazuh 4.14.7
Wazuh Server	Ubuntu Server 24.04 LTS
Windows Endpoint	Windows 11 Pro
Wazuh Manager IP	192.168.180.133
Windows Agent IP	192.168.180.131
Windows Agent ID	002

1. Objective

The objective of this lab was to deploy a small Wazuh SIEM environment, connect a Windows 11 endpoint using the Wazuh agent, generate controlled authentication activity, and investigate the resulting Windows security events in the Wazuh Threat Hunting interface.

The lab demonstrates an end-to-end SOC workflow: endpoint activity -> Windows event logs -> Wazuh agent -> Wazuh manager/indexer -> SIEM alert -> analyst investigation.

2. Lab Environment

The Wazuh all-in-one server was deployed on an Ubuntu Server 24.04 LTS virtual machine. The Wazuh manager, indexer, and dashboard were hosted on the same server. A separate Windows 11 Pro virtual machine was enrolled as the monitored endpoint.

Component	Configuration
Host platform	VMware Workstation
Wazuh server	Ubuntu Server 24.04 LTS
Wazuh manager/indexer/dashboard	192.168.180.133
Monitored endpoint	Windows 11 Pro
Wazuh agent	ID 002, DESKTOP-6CJNNAI
Endpoint address	192.168.180.131
Network mode	VMware NAT

3. Deployment and Agent Enrollment

The Wazuh server was installed using the all-in-one deployment. The Windows endpoint was then enrolled with the Wazuh agent and connected to the manager at 192.168.180.133.

Server-side verification confirmed that agent ID 002 was active. The Wazuh dashboard also showed the Windows endpoint as an active agent.

Verification result: Agent 002 / DESKTOP-6CJNNAI - Active

4. Test Activity

To generate authentication telemetry safely, the Windows 11 VM was locked and several incorrect password attempts were entered. The correct credentials were then used to unlock the workstation. This produced both failed and successful Windows authentication events for SIEM analysis.

5. Failed Logon Investigation - Event ID 4625

The Wazuh Threat Hunting dashboard was filtered for Windows Event ID 4625. Multiple authentication failures were visible, and an individual event was expanded for analysis.

Field	Observed value	Analyst interpretation
Event ID	4625	Windows logon failure
Agent ID	002	Monitored Windows endpoint
Agent IP	192.168.180.131	Windows 11 VM
Agent name	DESKTOP-6CJNNAI	Endpoint hostname
Rule description	Logon Failure - Unknown user or bad password	Wazuh authentication failure alert
Logon Type	2	Interactive/local logon attempt
Source IP	127.0.0.1	Localhost, not a remote source
Authentication package	Negotiate	Windows selected an available authentication method
Process	C:\Windows\System32\svchost.exe	Windows service host involved in event generation
Status	0xC000006D	Logon failure
Rule level	5	Moderate Wazuh alert severity
Fired times	4	Repeated failed attempts observed

The event was generated locally. Logon Type 2 and source address 127.0.0.1 are consistent with the controlled lock-screen authentication test. The evidence did not indicate a remote brute-force source.

Wazuh mapped this alert to MITRE ATT&CK; T1531 (Account Access Removal). This mapping was recorded as part of the alert metadata, but the lab evidence itself only demonstrated repeated local authentication failures. No account access removal activity was performed.

6. Successful Logon Investigation - Event ID 4624

The Wazuh Threat Hunting view was then filtered for Event ID 4624 to identify the successful authentication activity that followed the failed attempts.

Field	Observed value	Analyst interpretation
Event ID	4624	Successful Windows logon
Agent ID	002	Same monitored endpoint
Agent IP	192.168.180.131	Windows 11 VM
Agent name	DESKTOP-6CJNNAI	Endpoint hostname
Rule description	Non network or service local logon.	Wazuh local logon alert
Logon Type	7	Workstation unlock
Target domain	MicrosoftAccount	Microsoft account authentication context
Process	C:\Windows\System32\lsass.exe	Local Security Authority process
Authentication package	Negotiate	Windows authentication selection

Field	Observed value	Analyst interpretation
Severity	AUDIT_SUCCESS	Successful audit event
Rule level	3	Low-severity Wazuh alert

7. Event Comparison

Event	Result	Logon Type	Meaning
4625	Failure	2	Interactive/local logon attempt failed
4624	Success	7	Workstation unlocked successfully

8. Incident Interpretation

The sequence observed in Wazuh consisted of several failed local authentication attempts followed by a successful workstation unlock. The failed events were generated from localhost and were expected as part of the controlled test. No evidence of a remote source, malware execution, or account compromise was identified from the reviewed events.

From a SOC perspective, the investigation demonstrated how repeated authentication failures can be identified, filtered, and compared with later successful authentication activity to establish context and reduce false escalation.

9. Troubleshooting Performed

During the lab, the Wazuh indexer failed to start because of corrupted OpenSearch/Lucene index state left after an earlier failed installation and an unexpected host restart. The corrupted index data was isolated, the indexer security configuration was re-initialized, and the dashboard was restarted successfully.

A later unexpected host restart also caused filesystem inconsistencies on the Ubuntu VM. The root ext4 filesystem was repaired using fsck, after which Ubuntu and the Wazuh services returned to normal operation.

This troubleshooting provided additional practical experience with Linux service status checks, Wazuh/OpenSearch logs, filesystem recovery, and service recovery.

10. Key Skills Demonstrated

Skill	Evidence from lab
SIEM deployment	Installed and operated a Wazuh all-in-one server
Endpoint onboarding	Connected a Windows 11 Wazuh agent
Windows log analysis	Investigated Event IDs 4625 and 4624
Threat hunting	Filtered and reviewed alerts in Wazuh Threat Hunting
Authentication analysis	Compared failed interactive logons with successful workstation unlock
SOC triage	Distinguished controlled local activity from remote attack indicators
Linux administration	Checked services, logs, storage, and recovered a damaged filesystem
Troubleshooting	Recovered Wazuh indexer/dashboard availability after corruption

11. Conclusion

The lab successfully demonstrated end-to-end SIEM monitoring using Wazuh. A Windows 11 endpoint was enrolled, authentication events were collected, and both failed and successful logons were investigated in the SIEM. The analysis showed repeated local failed logon attempts followed by a successful workstation unlock, with no evidence of a remote brute-force source in the reviewed data.

The lab also included realistic troubleshooting of SIEM backend and Linux filesystem issues, strengthening practical SOC, SIEM, Windows event analysis, and Linux administration skills.

Portfolio summary: Deployed a Wazuh SIEM lab, onboarded a Windows 11 endpoint, generated and investigated Windows authentication events (4625/4624), analyzed source/logon context, and recovered Wazuh services after index and filesystem issues.